

EXP NO: 3
DATE:

Google App Engine SDK and Web App Development

Aim:

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

Pre-Requisites:

- Windows 10 Host Machine
- Oracle VM VirtualBox (Open Source)
- Ubuntu Desktop ISO image (Open Source)

Procedure:

Step 1: Install Google Cloud CLI & Python

Ensure Python 3 is installed on the system.

```
ubuntu@ubuntu:~$ sudo apt install python3
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
```

Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-
cloud-cli-linux-x86_64.tar.gz
tar -xf google-cloud-cli-linux-x86_64.tar.gz
./google-cloud-sdk/install.sh
```

Step 2: Initialize GCloud & Install Components

```
gcloud init
```

Sign in to your Google account when prompted.

```

lakshya@lakshya-VirtualBox:~$ gcloud init
Welcome! This command will take you through the configuration of gcloud.

Your current configuration has been set to: [default]

You can skip diagnostics next time by using the following flag:
  gcloud init --skip-diagnostics

Network diagnostic detects and fixes local network connection issues.
Checking network connection...done.
Reachability Check passed.
Network diagnostic passed (1/1 checks passed).

You must sign in to continue. Would you like to sign in (selecting "Y" will open your browser to the sign-in page where you
complete authentication)
(Y/n)? y

Your browser has been opened to visit:
  https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=http%3A%2F%2Flocalhost%3A8080%2F&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&state=LbPLNpuKjLz8QztLCeM1AowZAatnp&access_type=offline&code_challenge=V5izKItNj1ML_Kc
m0xp8tJ8IS3x59QPCuFP_IFLE&code_challenge_method=S256

```

gcloud components install app-engine-python

```

lakshya@lakshya-VirtualBox:~$ gcloud components install app-engine-python

Your current Google Cloud CLI version is: 574.0.0
Installing components from version: 574.0.0

These components will be installed.


| Name                                    | Version | Size     |
|-----------------------------------------|---------|----------|
| Cloud Datastore Emulator                | 2.3.1   | 36.2 MiB |
| gRPC Python library (Platform Specific) | 1.70.0  | 5.6 MiB  |
| gcloud app Python Extensions            | 1.9.118 | 3.8 MiB  |



For the latest full release notes, please visit:
https://cloud.google.com/sdk/release\_notes

Once started, canceling this operation may leave your SDK
installation in an inconsistent state.

Do you want to continue (Y/n)? Y

Performing in place update...

```

Performing in place update...
Downloading: Cloud Datastore Emulator
Downloading: gRPC Python library
Downloading: gRPC Python library (Platform Specific)
Downloading: gcloud app Python Extensions
Installing: Cloud Datastore Emulator
Installing: gRPC Python library
Installing: gRPC Python library (Platform Specific)
Installing: gcloud app Python Extensions
Performing post processing steps...done.
Update done!

Step 3: Create the App Directory

```

mkdir gae-hello-world
cd gae-hello-world

```

Step 4: Create Application Code (main.py)

```

from flask import Flask
app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello World from Google App Engine!"

if __name__ == "__main__":
    app.run(host="127.0.0.1", port=8080, debug=True)

```

Step 5: Create Configuration File (app.yaml)

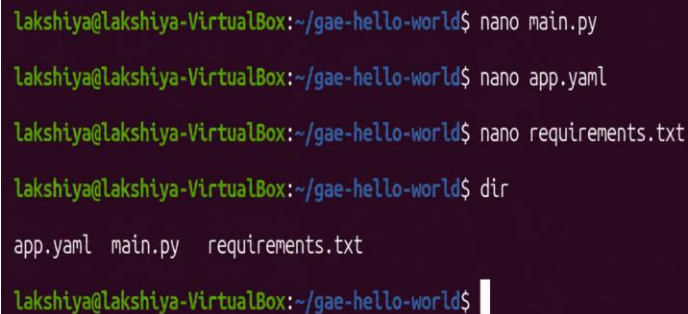
```
runtime: python39
handlers:
- url: /*
  script: auto
```

Step 6: Create requirements.txt

```
Flask==2.0.1
```

Output:

Files created in project directory (main.py, app.yaml, requirements.txt):

A terminal window with a dark purple background and green text. It shows a series of commands and their outputs. The commands are: 'nano main.py', 'nano app.yaml', 'nano requirements.txt', and 'dir'. The output of 'dir' is 'app.yaml main.py requirements.txt'.

```
lakshiya@lakshiya-VirtualBox:~/gae-hello-world$ nano main.py
lakshiya@lakshiya-VirtualBox:~/gae-hello-world$ nano app.yaml
lakshiya@lakshiya-VirtualBox:~/gae-hello-world$ nano requirements.txt
lakshiya@lakshiya-VirtualBox:~/gae-hello-world$ dir
app.yaml main.py requirements.txt
lakshiya@lakshiya-VirtualBox:~/gae-hello-world$
```

Result:

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.